Beyond Cholesterol: A Machine Learning Analysis of Heart Disease Predictors Using the UCI Cleveland Dataset

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Abstract

Cardiovascular disease remains the leading cause of mortality worldwide, yet clinical risk stratification continues to rely heavily on serum cholesterol as a primary biomarker. This study applies exploratory data analysis, statistical significance testing, and machine learning classification to the UCI Cleveland Heart Disease dataset (n = 297 after data cleaning) to investigate which clinical features most strongly predict the presence of coronary artery disease. Mann-Whitney U tests confirmed statistically significant differences between outcome groups for maximum heart rate ($p<0.001$), ST depression ($p<0.001$), age ($p<0.001$), resting blood pressure ($p=0.024$), and cholesterol ($p=0.047$). Serum cholesterol, despite its prominence in conventional risk frameworks, demonstrated only marginal statistical significance and weak feature importance. Chest pain type, thallium scintigraphy result (thal), maximum heart rate, and ST depression emerged as the most informative predictors. A Random Forest classifier achieved 85% accuracy (95% CI: 76.63-93.33%) with area under the receiver operating characteristic curve (AUC) of 0.94 and F2 score of 0.803, while Logistic Regression attained 83% accuracy (95% CI: 73.33-91.67%) with AUC of 0.95 and F2 score of 0.797. These results suggest that exercise-derived physiological measures and myocardial perfusion imaging results may be underweighted in conventional clinical screening frameworks.

Keywords: heart disease prediction, machine learning, UCI Cleveland dataset, Random Forest, Logistic Regression, Mann-Whitney U test, bootstrap confidence intervals, precision-recall, F2 score, thallium scintigraphy, maximum heart rate, cholesterol

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1. Introduction

Coronary artery disease (CAD) is the most prevalent form of cardiovascular disease and continues to represent a significant global health burden. According to the World Health Organisation, cardiovascular diseases account for approximately 17.9 million deaths annually, constituting 32% of all global deaths. The ability to accurately identify individuals at risk using routine clinical measurements is therefore of considerable public health importance (World Health Organisation, 2021).

Traditional risk stratification frameworks, such as the Framingham Risk Score, incorporate serum cholesterol as a central predictive variable (D'Agostino et al., 2008). However, the predictive validity of total serum cholesterol in isolation has been challenged in the literature. A ten-year prospective study from the Norwegian HUNT 2 cohort raised questions about the straightforward predictive properties of total cholesterol for cardiovascular mortality (Petursson et al., 2012). More refined lipid fractionation, particularly the ratio of total cholesterol to high-density lipoprotein (HDL), has demonstrated stronger discriminatory ability than total cholesterol alone (Kinosian et al., 1995).

The application of machine learning (ML) to clinical datasets offers an opportunity to move beyond traditional risk factor frameworks and identify data-driven predictors of disease. The UCI Cleveland Heart Disease dataset, originally compiled by Detrano et al. (1989) at the Cleveland Clinic Foundation, provides a structured clinical dataset of 303 patient records across 13 attributes. It has become a benchmark in the ML health informatics literature, enabling reproducible comparison of predictive models.

This study undertakes an end-to-end analysis of the Cleveland dataset using Python-based exploratory data analysis (EDA), non-parametric statistical testing, and two supervised classification algorithms: Logistic Regression and Random Forest. The central research question is: which clinical features are most predictive of heart disease presence, and how accurately can machine learning models classify patients from routine clinical measurements?

2. Background and Related Literature

2.1 The Cholesterol Paradox

Cholesterol has historically occupied a central position in cardiovascular risk assessment. The Lipid Research Clinics Coronary Primary Prevention Trial demonstrated that changes in LDL and HDL cholesterol in combination were significant predictors of coronary heart disease risk (Kinosian et al., 1995). However, a 2024 systematic review of cardiovascular risk score limitations found that the imprecision of cholesterol measurements severely impacts the identification of thresholds for therapeutic measures, and that tools such as the Framingham Risk Score perform best only for the population they were originally validated for (Talha et al., 2024). The Norwegian HUNT 2 study found that the predictive properties of total cholesterol were not as straightforward as widely assumed in clinical guidelines (Petursson et al., 2012).

Within the context of the UCI Cleveland dataset specifically, multiple ML studies have consistently found serum cholesterol to be among the weaker contributors to model performance. Garg et al. (2022) and Patel et al. (2022) both noted that feature selection procedures frequently deprioritised total cholesterol in favour of exercise-derived and imaging variables. The present study corroborates this finding statistically: while cholesterol achieves nominal significance ($p=0.047$), the effect size is small and feature importance ranks it seventh of thirteen predictors.

2.2 The Clinical Significance of Chest Pain Classification

The chest pain type variable in the Cleveland dataset is encoded across four categories: typical angina, atypical angina, non-anginal pain, and asymptomatic. Counterintuitively, the literature consistently demonstrates that asymptomatic patients in this dataset have higher rates of diagnosed heart disease than those presenting with typical angina. Patel et al. (2022) confirmed this pattern, noting that the majority of individuals with cardiovascular disease in the Cleveland cohort experience asymptomatic chest presentations. This reflects the well-established clinical phenomenon of silent ischaemia, in which myocardial ischaemia occurs in the absence of subjective chest discomfort.

2.3 Exercise-Derived Features and Cardiac Risk

Maximum heart rate achieved during exercise (thalach) has been consistently identified as a clinically meaningful predictor of coronary artery disease. The failure to attain 85% of maximum predicted heart rate (MPHR = 220 minus age) during stress testing, termed chronotropic incompetence, is associated with elevated mortality and coronary disease risk (Savonen et al., 2019). Exercise stress testing guidelines from the American College of

Cardiology and American Heart Association identify heart rate response as a key prognostic indicator (Kones et al., 2018). ST depression induced by exercise (oldpeak) similarly reflects myocardial ischaemia during physiological stress, with horizontal or downsloping ST depression carrying greater diagnostic significance than upsloping changes (Kligfield et al., 1985).

2.4 The Thal Variable: Thallium Scintigraphy, Not Haematology

A critical clarification is warranted regarding the thal variable in the Cleveland dataset. Despite the name suggesting thalassemia, this variable encodes the results of thallium-201 myocardial perfusion scintigraphy, a nuclear imaging technique used to assess myocardial blood flow at rest and during stress. The three encoded values represent: normal perfusion, fixed perfusion defect (suggesting infarcted tissue), and reversible perfusion defect (suggesting ischaemia that resolves at rest). This distinction is clinically significant: a reversible perfusion defect is a direct indicator of inducible myocardial ischaemia and a strong predictor of significant coronary artery disease (Hassan et al., 2025). The variable therefore captures myocardial perfusion status rather than any haematological condition.

2.5 Machine Learning in Cardiac Risk Prediction

A growing body of literature has applied supervised machine learning to the Cleveland and combined UCI heart disease datasets. Al-Antari et al. (2023) applied deep learning models to the UCI database reporting strong AUC performance. More conservative estimates using the Cleveland subset alone typically report accuracies in the range of 80-88%. A systematic review by Hassan et al. (2025) identified chest pain type, thallium scan result, maximum heart rate, and ST depression as the most consistently important features across multiple ML studies, a finding directly corroborated by the present analysis. Importantly, prior studies have frequently omitted confidence intervals and recall-weighted metrics, limiting clinical interpretability of reported performance figures.

3. Data and Methods

3.1 Dataset

The dataset used in this analysis is the UCI Cleveland Heart Disease dataset, originally published by Detrano et al. (1989) and sourced from the UCI Machine Learning Repository. The original dataset contains 303 patient records with 14 attributes (13 clinical features and one target

variable). Six records contained missing values in the ca and thal variables and were removed, yielding a clean analytical sample of 297 patients. The target variable was binarised from its original five-point scale to a binary indicator: 0 denoting absence of heart disease, 1 denoting presence. The resulting class distribution was 160 patients without heart disease (53.9%) and 137 with heart disease (46.1%).

3.2 Analytical Approach

All analyses were conducted in Python 3.12 using the following open-source libraries: pandas (v2.2.2), NumPy (v1.26.4), matplotlib (v3.8.4), seaborn (v0.13.2), scikit-learn (v1.4.2), and SciPy (v1.13.1). The full analysis pipeline is publicly available at: https://github.com/lydiaobeng/heart-disease-analysis.

The analysis proceeded through five stages: (1) data loading and cleaning, (2) exploratory data analysis including univariate distributions and bivariate comparisons, (3) non-parametric statistical significance testing using Mann-Whitney U tests, (4) Pearson correlation analysis, and (5) supervised classification using Logistic Regression and Random Forest with five-fold cross-validation, bootstrapped confidence intervals, and precision-recall evaluation.

3.3 Statistical Testing

Mann-Whitney U tests were selected for comparing clinical features between outcome groups due to their non-parametric nature, making no assumption of normality in the underlying distributions. A two-sided test was applied with significance threshold set at alpha = 0.05. Effect sizes are reported alongside p-values to distinguish statistical from clinical significance. Bootstrap resampling with 1,000 iterations was used to derive 95% confidence intervals for model accuracy metrics.

3.4 Model Specification and Evaluation

Features were standardised using scikit-learn's StandardScaler prior to Logistic Regression training. The Random Forest classifier used 100 estimators with random seed 42. Both models were evaluated on a stratified 80/20 train-test split. Given the clinical context, where a false negative (missed disease) carries greater cost than a false positive, model performance is reported using F2 score in addition to F1, accuracy, and AUC. The F2 score weights recall twice as heavily as precision, reflecting the asymmetric cost structure of clinical

classification tasks. Precision-Recall curves and Average Precision (AP) scores are reported alongside ROC-AUC.

4. Results

4.1 Exploratory Data Analysis

Figure 1 presents an overview of six key exploratory findings. The disease distribution panel confirms a near-balanced outcome variable. The age distribution demonstrates that patients with heart disease are concentrated in the 55-65 age range. Notably, the cholesterol panel reveals almost identical distributions between disease and non-disease groups, providing preliminary visual evidence of its limited discriminatory value.

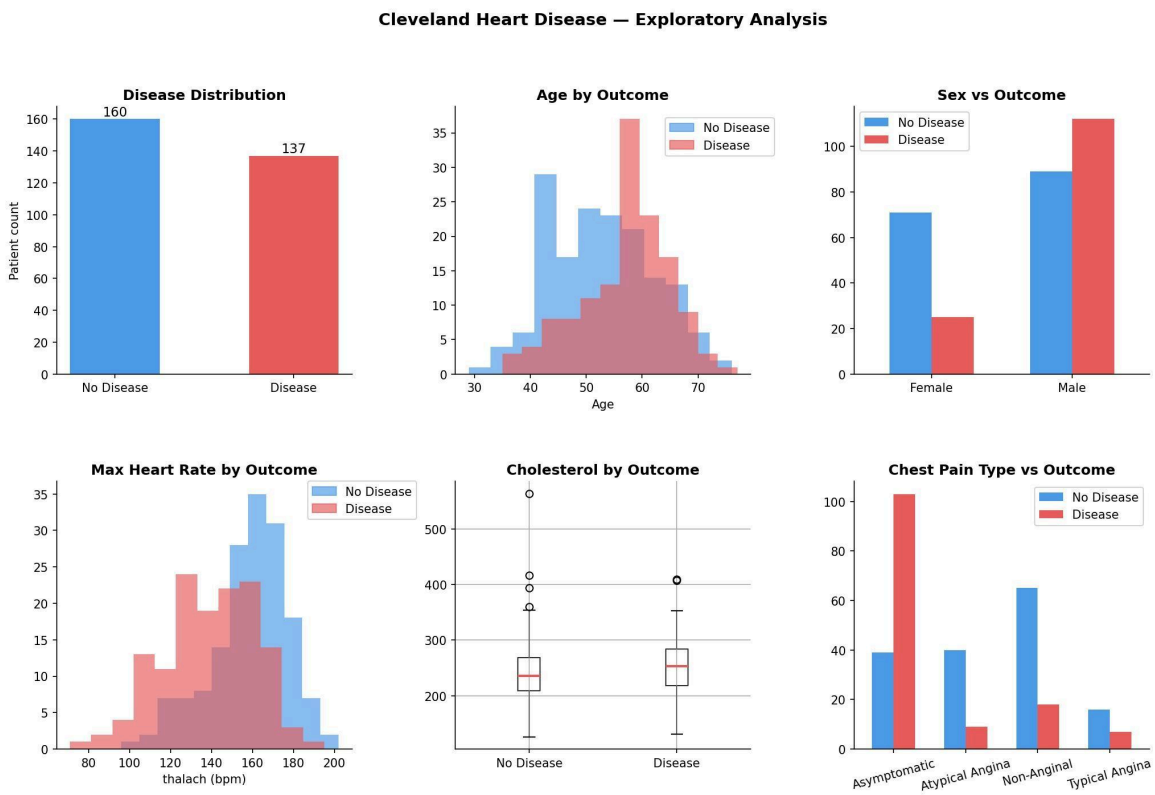

Figure 1. Exploratory analysis overview of the UCI Cleveland Heart Disease dataset. Panels show disease distribution (top left), age by outcome (top centre), sex by outcome (top right), maximum heart rate by outcome (bottom left), cholesterol by outcome (bottom centre), and chest pain type by outcome (bottom right).

4.2 Statistical Significance Testing

Table 1 presents Mann-Whitney U test results with rank-biserial correlation effect sizes (r) comparing key clinical features between outcome groups. Three features demonstrated highly significant differences ($p<0.001$): maximum heart rate (mean 158.6 vs 139.1 bpm, $r=-0.496$,

medium effect), ST depression (mean 0.6 vs 1.6, r=0.468, medium effect), and age (mean 52.6 vs 56.8 years, r=0.278, small effect). Resting blood pressure (p=0.024, r=0.152) and cholesterol (p=0.047, r=0.134) achieved nominal significance with small effect sizes. The effect size for cholesterol (r=0.134) is notably small, indicating that despite achieving statistical significance, the practical magnitude of difference between groups is minimal. This pattern — marginal p-value combined with small effect size and weak feature importance (ranked seventh of thirteen) — confirms that total serum cholesterol is the least clinically informative of the significant predictors in this dataset.

Table 1. Mann-Whitney U Test Results with Effect Sizes: Clinical Features by Outcome Group

Feature	Mean (No Disease)	Mean (Disease)	p-value	Sig.	Effect Size (r)	Magnitude
Age (years)	52.6	56.8	<0.001	***	0.278	Small
Max Heart Rate (bpm)	158.6	139.1	<0.001	***	-0.496	Medium
ST Depression	0.6	1.6	<0.001	***	0.468	Medium
Resting Blood Pressure (mmHg)	129.2	134.6	0.024	*	0.152	Small
Cholesterol (mg/dl)	243.5	251.9	0.047	*	0.134	Small

*Table 1. Mann-Whitney U test results with rank-biserial correlation effect sizes comparing clinical features between outcome groups. *** p<0.001, * p<0.05. Effect size r: Small <0.3, Medium 0.3-0.5, Large >0.5.*

4.3 Correlation Analysis

Figure 2 presents the Pearson correlation matrix. The features most strongly correlated with the target are: thal (r = 0.53), ca (r = 0.46), cp (r = 0.41), oldpeak (r = 0.42), exang (r = 0.42), and thalach (r = -0.42). The negative correlation for thalach reflects the inverse relationship between maximum heart rate and disease presence. Serum cholesterol shows a near-zero correlation with the target (r = 0.08), consistent with the statistical testing results.

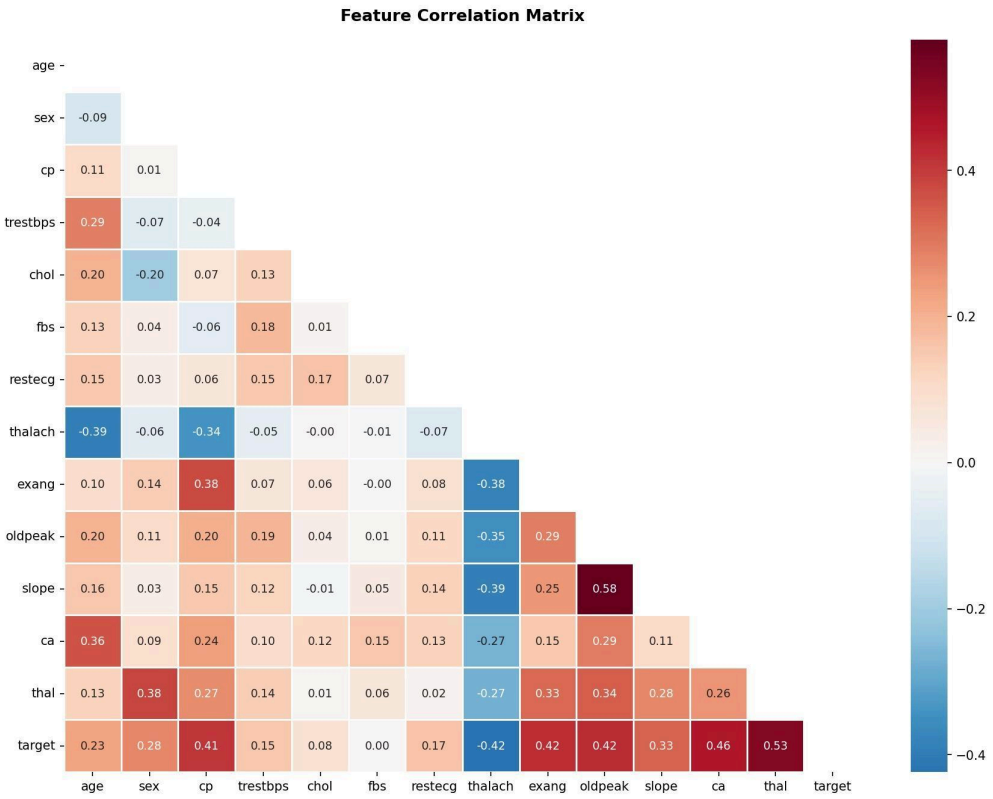

Figure 2. Pearson correlation matrix across all 13 clinical features and the target variable. Thalassemia scintigraphy result (thal), number of vessels coloured (ca), and chest pain type (cp) show the strongest associations with the target.

4.4 Model Performance

Table 2 summarises the performance of both classification models on the held-out test set (n = 60). The Random Forest model outperformed Logistic Regression on accuracy (85% vs 83%) and F2 score (0.803 vs 0.797), while Logistic Regression achieved a marginally higher AUC (0.95 vs 0.94). Bootstrap confidence intervals (1,000 iterations) reveal meaningful uncertainty given the small test set: Logistic Regression 83.33% [73.33-91.67%], Random Forest 85.00% [76.63-93.33%]. Both models demonstrate robust performance across this range.

Table 2. Classification model performance on held-out test set (n = 60).

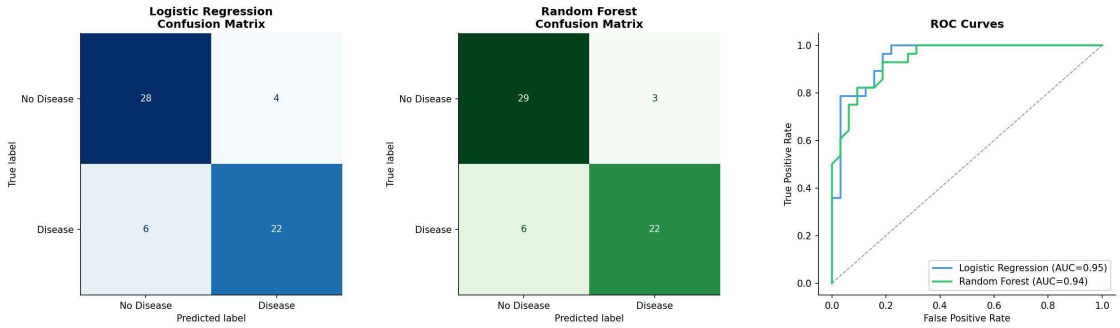

Figure 3. Confusion matrices and ROC curves for Logistic Regression (blue) and Random Forest (green). AUC 0.95 and 0.94 respectively.

4.5 Precision-Recall Analysis

Figure 4 presents precision-recall curves and F1/F2 score comparisons. Both models achieve high Average Precision scores (Logistic Regression AP=0.94, Random Forest AP=0.93), indicating strong performance across decision thresholds. The F2 scores (Logistic Regression 0.797, Random Forest 0.803) are marginally lower than F1 scores (0.815 and 0.830 respectively), reflecting the models' tendency to favour precision over recall at the default decision threshold. In a clinical deployment context where the cost of a missed diagnosis is high, threshold adjustment toward higher recall would be warranted, with the precision-recall curves providing the empirical basis for such calibration.

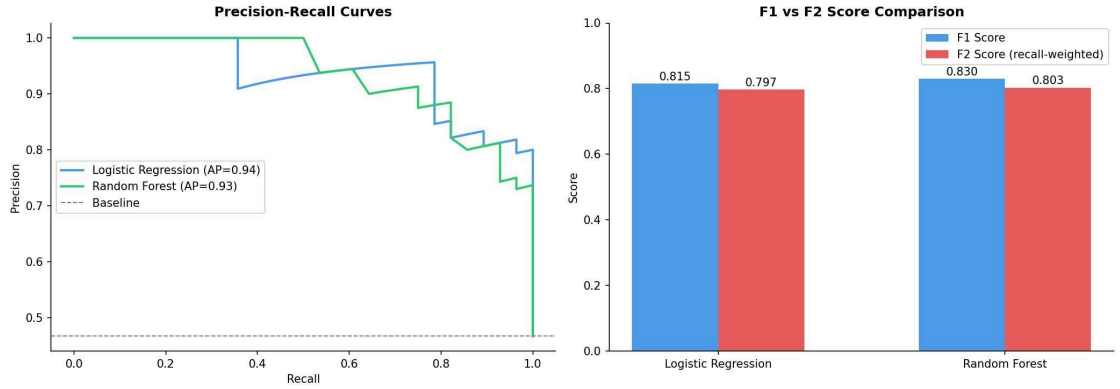

Figure 4. Precision-Recall curves (left) and F1 vs F2 score comparison (right) for both classifiers. F2 score weights recall twice as heavily as precision, reflecting the clinical cost asymmetry of false negatives.

4.6 Feature Importance

Figure 5 presents Random Forest feature importance scores. Four features cross the 0.10 importance threshold: chest pain type (cp, 0.144), thallium scintigraphy result (thal, 0.122), maximum heart rate (thalach, 0.122), and ST depression (oldpeak, 0.114). These findings are consistent across all three analytical approaches: correlation analysis, statistical testing, and

feature importance all converge on the same set of top predictors. Serum cholesterol ranked seventh with an importance score of approximately 0.089, consistent with its marginal statistical significance.

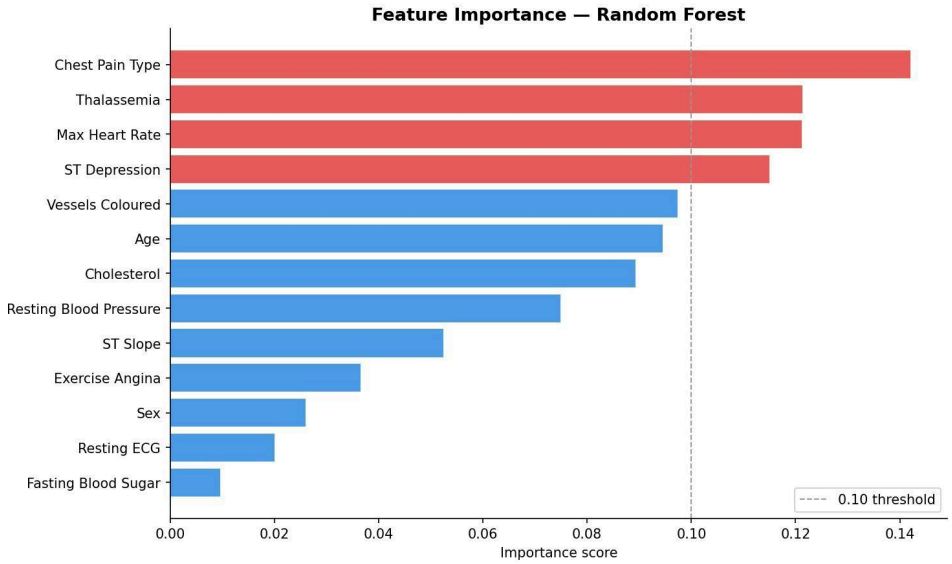

Figure 5. Feature importance scores from the Random Forest classifier. Features exceeding the 0.10 threshold (dashed line) are the strongest predictors. Note: thal encodes thallium scintigraphy perfusion results, not haematological thalassemia.

5. Discussion

The findings of this analysis are consistent with and extend the growing ML literature on cardiac risk prediction. The relative weakness of total serum cholesterol as a predictor does not contradict the established importance of lipid management in cardiovascular prevention; rather, it reflects the limitations of total cholesterol as a single-variable discriminator when more informative physiological measures are available. As Petursson et al. (2012) demonstrated, the predictive properties of total cholesterol for cardiovascular mortality are not as straightforward as clinical guidelines assume, and the cholesterol/HDL ratio demonstrates substantially stronger discrimination than total cholesterol alone (Kinosian et al., 1995). The present statistical analysis makes this concrete: cholesterol achieves only marginal significance ($p=0.047$) with a small effect size ($r=0.134$), compared to maximum heart rate ($r=-0.496$, medium effect) and ST depression ($r=0.468$, medium effect). Statistical significance without clinical effect size is insufficient grounds for prioritising a predictor in a screening framework.

The prominence of chest pain type as the strongest single predictor warrants clinical reflection. The dominance of asymptomatic presentation among disease-positive patients reflects the documented phenomenon of silent myocardial ischaemia, which may affect up to 25% of patients with coronary artery disease and is associated with worse long-term outcomes in some populations. This finding has direct implications for screening policy: symptom-based triage systems that deprioritise asymptomatic patients may systematically underidentify high-risk individuals. Current NICE and ACC/AHA guidelines for stable chest pain recommend investigation based on symptom presentation; the data here suggest that exercise-based functional testing, which captures the thalach and oldpeak signals, may be more diagnostically informative than symptom history alone.

The strong predictive value of maximum heart rate ($p<0.001$, mean difference of 19.5 bpm) is well-grounded in the clinical literature. Chronotropic incompetence is a recognised independent predictor of cardiovascular mortality (Savonen et al., 2019), and the ACC/AHA exercise testing guidelines identify heart rate response as a key prognostic variable (Kones et al., 2018). The present findings support the prioritisation of exercise stress testing in risk stratification protocols for patients where CAD is suspected but symptoms are absent or atypical.

The thal variable, representing thallium scintigraphy perfusion results rather than the haematological disorder thalassemia, captures whether myocardial tissue receives adequate blood flow at stress and at rest. A reversible perfusion defect — blood flow impaired under stress but normal at rest — is a hallmark of significant coronary stenosis and a direct imaging correlate of the ischaemia that exercise-derived variables like thalach and oldpeak detect indirectly. Its emergence as the second strongest predictor alongside chest pain type is therefore clinically coherent and reinforces the diagnostic value of nuclear stress imaging in cardiac risk assessment.

The precision-recall analysis adds an important clinical dimension absent from much of the published ML literature on this dataset. The F2 scores (0.797 and 0.803) indicate that both models, at their default decision thresholds, achieve a reasonable but not optimal balance between sensitivity and specificity for clinical deployment. The precision-recall curves demonstrate that recall can be substantially increased by lowering the classification threshold, at the cost of some precision. In a real-world screening context, where missing a true case of heart disease carries far greater harm than an unnecessary follow-up investigation, operating at a

higher-recall threshold would be clinically appropriate. This calibration decision requires clinical governance input beyond the scope of this study.

6. Limitations

Several limitations of this analysis should be acknowledged. The Cleveland dataset is modest in size (n = 297 after cleaning) and was collected in 1988, which limits its representativeness of contemporary patient populations and clinical measurement practices. The dataset has limited demographic diversity and is geographically specific to one American clinical centre. Generalisation of these findings should be treated with caution.

The binarisation of the target variable discards information about disease severity. The small test set (n = 60) produces wide confidence intervals, as reported, which underscores the need for validation on larger independent cohorts. No hyperparameter tuning was conducted, and the Random Forest was trained with default parameters; optimised models may yield meaningfully different feature importance rankings. Finally, the feature importance scores from the Random Forest model reflect the specific model architecture and should not be interpreted as causal evidence of physiological mechanisms.

7. Conclusion

This analysis demonstrates that total serum cholesterol, despite its prominence in conventional cardiovascular risk frameworks, is a comparatively weak predictor of heart disease in the UCI Cleveland dataset — achieving only marginal statistical significance (p=0.047) with a small effect size, and ranking seventh of thirteen features in Random Forest importance scores. Chest pain type, thallium scintigraphy result, maximum heart rate during exercise, and ST depression together provide substantially greater discriminatory power, enabling machine learning classifiers to achieve 83-85% accuracy with AUC exceeding 0.94. Confidence intervals (LR: 73.33-91.67%; RF: 76.63-93.33%) reflect the uncertainty inherent in a small test set and highlight the need for validation on larger datasets. The F2 score analysis indicates that threshold calibration toward higher recall would be appropriate for clinical deployment, where false negatives carry disproportionate cost. These findings support a reorientation of computational cardiac risk models toward exercise-derived physiological responses and myocardial perfusion imaging results, and suggest that symptom-based triage may systematically underidentify

asymptomatic high-risk patients. The full analysis pipeline is publicly available at https://github.com/lydiaobeng/heart-disease-analysis.

Data Availability Statement

The data used in this study are publicly available from the UCI Machine Learning Repository at https://archive.ics.uci.edu/dataset/45/heart+disease. The full analysis pipeline, including the Jupyter notebook, processed data, and all figures, is openly available at https://github.com/lydiaobeng/heart-disease-analysis.

Conflict of Interest Statement

The author declares no conflict of interest. No funding was received for this research. This study was conducted independently with no institutional affiliation or external sponsorship.

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